

A fine crop classification model based on multitemporal Sentinel-2 images

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ABSTRACT

Information on the sowing areas and yields of crops is important for ensuring food security and reforming the agricultural modernization process, while crop classification and identification are core issues when attempting to acquire information on crop planting areas and yields. Obtaining information on crop planting areas and yields in a timely and accurate manner is highly important for optimizing crop planting structures, formulating agricultural policies, and ensuring national economic development. In this paper, a fine crop classification model based on multitemporal Sentinel-2 images, CTANet, is proposed. It comprises a convolutional attention architecture (CAA) and a temporal attention architecture (TAA), incorporating spatial attention modules, channel attention modules and temporal attention modules. These modules adaptively weight each pixel, channel and temporal phase of the given feature map to mitigate the intraclass spatial heterogeneity, spectral variability and temporal variability of crops. Additionally, the auxiliary features of significant importance for each crop category are identified using the random forest-SHAP algorithm, enabling the construction of classification datasets containing spectral bands, spectral bands with auxiliary features, and spectral bands with optimized auxiliary features. Evaluations conducted on three crop classification datasets revealed that the proposed CTANet approach and its key CANet component demonstrated superior crop classification performance on the classification dataset consisting of spectral bands and optimized auxiliary features in comparisons with the other tested models. Based on this dataset, CTANet achieved higher validation accuracy and lower validation loss than those of the other methods, and during testing, it attained the highest overall accuracy (93.9 %) and MIoU (87.5 %). When identifying rice, maize, and soybeans, the F1 scores of CTANet reached 95.6 %, 95.7 %, and 94.7 %, and the IoU scores were 91.6 %, 91.7 %, and 89.9 %, respectively, significantly exceeding those of some commonly used deep learning models. This indicates the potential of the proposed method for distinguishing between different crop types.

1. Introduction

Information on the sowing areas and yields of crops is important for ensuring food security and reforming the agricultural modernization process (Karhikeyan et al., 2020; Su and Zhang, 2023), while crop classification and identification are core issues when attempting to acquire information on crop planting areas and yields (Blickensdorfer et al., 2022). Traditional methods for surveying crop planting conditions mainly rely on agricultural statistical reports submitted at various levels and catalog sampling based on statistics, employing human resources to investigate crop types, locations, and other field attributes. This method not only is time-consuming and labor-intensive but also struggles to

provide real-time information concerning the geographical distributions of crops, failing to satisfy the current societal demands for the rapid acquisition and management of agricultural production information (Sonobe, 2019; Heping Li et al., 2022). Therefore, acquisition information on crop planting areas and yields in an accurate and timely manner is highly important for optimizing crop planting structures, scientifically formulating agricultural policies, and ensuring national economic development.

Since the beginning of the 21st century, spatial Earth information acquisition technology has rapidly developed, and mature satellite-based, airborne, and ground-based remote sensing platforms have been established (Ghamisi et al., 2019). With the continuous

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advancement of imaging technology involving various remote sensing devices, sustained improvements in the spatial, temporal, and spectral resolutions of remote sensing data have been observed (Seydi et al., 2021b), providing more detailed and diverse data sources for monitoring the same region. Remote sensing observations, which cover various spatial resolutions and scales, are widely used for crop mapping (Azzari and Lobell, 2017; Xu et al., 2020). From the 1970s to the present, many countries have conducted a series of studies on remote sensing-based crop area monitoring. Two typical national-scale applications are the Cropland Data Layer (CDL) implemented by the United States Department of Agriculture-National Agricultural Statistics Service (USDA-NASS) and the Crop Inventory (CI) released by Agriculture and Agri-Food Canada (AAFC) (Boryan et al., 2011; Fiset et al., 2013). These programs can provide annual spatial distribution maps for major crops across their respective countries. Monitoring agricultural production activities using remote sensing enables the accurate and efficient transmission of crucial information at various stages to agricultural production managers, assisting them in making relevant decisions and formulating plans (Lira Melo de Oliveira Santos et al., 2019). This approach optimizes resource allocation, enhances production efficiency, and effectively addresses potential risks and challenges, ensuring the sustainability and stability of agricultural production.

The key to using remote sensing technology for agricultural monitoring lies in precisely identifying various crop types. Traditional crop classification methods based on remote sensing imagery can be divided into pixel-based classification methods and object-oriented classification methods according to their processing units (Belgiu and Csillik, 2018; Ghamisi et al., 2019; Su and Zhang, 2023). The former approaches primarily rely on spectral features, while the latter methods mainly depend on spectral, shape, textural, and other features. However, these learning approaches focus only on surface features; when addressing deeper and more complex patterns and features, further manual data processing is needed, and the learning effect is not strong. Crop classification tasks cannot maximize or deeply leverage the information inherent in the data themselves. With the explosive growth of remote sensing data, as well as the resolution, diversity, and complexity enhancements exhibited by remote sensing images, shallow models are becoming inadequate for satisfying the current image processing needs, especially when faced with a vast amount of high-resolution, multi-modal, and multitemporal remote sensing data. Compared to traditional models, deep learning models possess multilayer network structures that can automatically learn hierarchical features from the data without the need for manual design and feature extraction strategies. This approach provides deep learning models with a significant advantage when handling complex remote sensing data, and in recent years, they have become advanced remote sensing classification methods (Seydi et al., 2021a).

Based on the three main information dimensions of spatial, spectral, and temporal data from remote sensing images, current deep learning models for crop classification can be categorized into four types. The first type is models designed for learning spatial-spectral features, typically using single-temporal images as input data and processing them with two-dimensional convolutional neural networks (2D CNNs) or pre-trained deep network structures such as ResNet and VGG (Kwak and Park, 2022; L. Wang et al., 2022). The second type is models designed for learning spectral features only, which often utilize multiple spectral bands of the image and process them using fully connected networks or one-dimensional convolutional neural networks (1D CNNs) (Khotimah et al., 2022; Mao et al., 2023). The third type is models designed for learning spectral-temporal features, employing multi-temporal images or extracted vegetation index time series as input and performing classification with 1D CNNs or recurrent neural networks (RNNs) such as LSTM and GRU (Ndikumana et al., 2018; Yi et al., 2022). The fourth type is models designed for comprehensive learning of spatial-spectral-temporal features, which can be approached in three ways: constructing two convolutional neural networks based on multi-

temporal images or extracted vegetation index time series to extract spatial-spectral and spectral-temporal features respectively (Seydi et al., 2022; Y. Wang et al., 2022); building datasets for each temporal instance and using three-dimensional convolutional neural networks (3D CNNs) combined with fully connected networks or solely four-dimensional convolutional neural networks (4D CNNs) for deep integrated learning of the three main information dimensions (Ji et al., 2018; Giannopoulos et al., 2022); or combining 2D CNNs with RNNs or Markov processes to extract spatial-spectral-temporal features from temporal datasets (Khaki et al., 2020; Huapeng Li et al., 2022). Existing crop classification models have certain limitations when handling spatio-temporal-spectral features. For instance, traditional CNNs cannot effectively capture temporal variations, traditional RNNs can only process temporal sequence features and are incapable of learning spatial features. Additionally, models that comprehensively learn spatial-spectral-temporal features, despite their powerful learning capabilities, suffer from large parameter sizes, low training efficiency, and poor performance on small sample datasets.

To overcome these limitations, this study aims to develop a deep learning model that can deeply explore crop features in the spatial, channel, and temporal dimensions under general hardware conditions and with small sample datasets. The objective is to achieve precise crop identification and spatial distribution mapping. The primary study content is divided into three parts. First, feature selection is performed using the random forest-SHAP (RF-SHAP) algorithm to evaluate the importance of shallow features (such as vegetation indices and textural features) for remote sensing crop classification, selecting features with high importance, and constructing a classification dataset consisting of optimized auxiliary features. Second, the model construction process combines a convolutional attention architecture with a temporal attention architecture to establish a hybrid neural network model called CTANet for the deep mining of temporal-spatial-spectral features. Finally, the overall accuracy, user accuracy, producer accuracy, F1 score, IoU, and MIOU metric are used to evaluate and analyze the classification performance of CTANet and the current mainstream deep learning models. The main contributions and innovations of this study are as follows:

- (1) Proposed a feature fusion strategy that integrates selected auxiliary features with spectral bands into a semantic segmentation model, and investigated the impact of different feature combinations on crop classification.
- (2) Designed two architectures tailored for small-sample datasets, adapting to a dataset with only 268 labeled images, effectively leveraging limited data to enhance model performance.
- (3) Introduced multi-dimensional attention mechanisms to enable the model to comprehensively capture spatial, spectral, and temporal features when processing multi-temporal remote sensing images, and explored the influence of these attention mechanisms on crop classification.

2. Study area and datasets

2.1. Study area

The study area is Youyi County (Fig. 1), which is situated in the northeastern part of Heilongjiang Province, China, at the heart of the Sanjiang Plain. It stretches longitudinally from 46°31'N to 46°59'N and latitudinally from 131°28'E to 132°15'E. It shares its eastern and northern borders with Fujin City, faces Baoqing County across the Qixing River to the south, borders Baoshan District to the southwest, and is adjacent to Jixian County to the west. Covering an area of 1,696 square kilometers, Youyi County is recognized as one of China's key grain-producing bases and a showcase for modernized agriculture. It has earned the title "Granary of the Northeast." The county boasts a temperate continental climate characterized by moderate temperatures, fertile soil, and abundant sunlight. The annual average temperature is

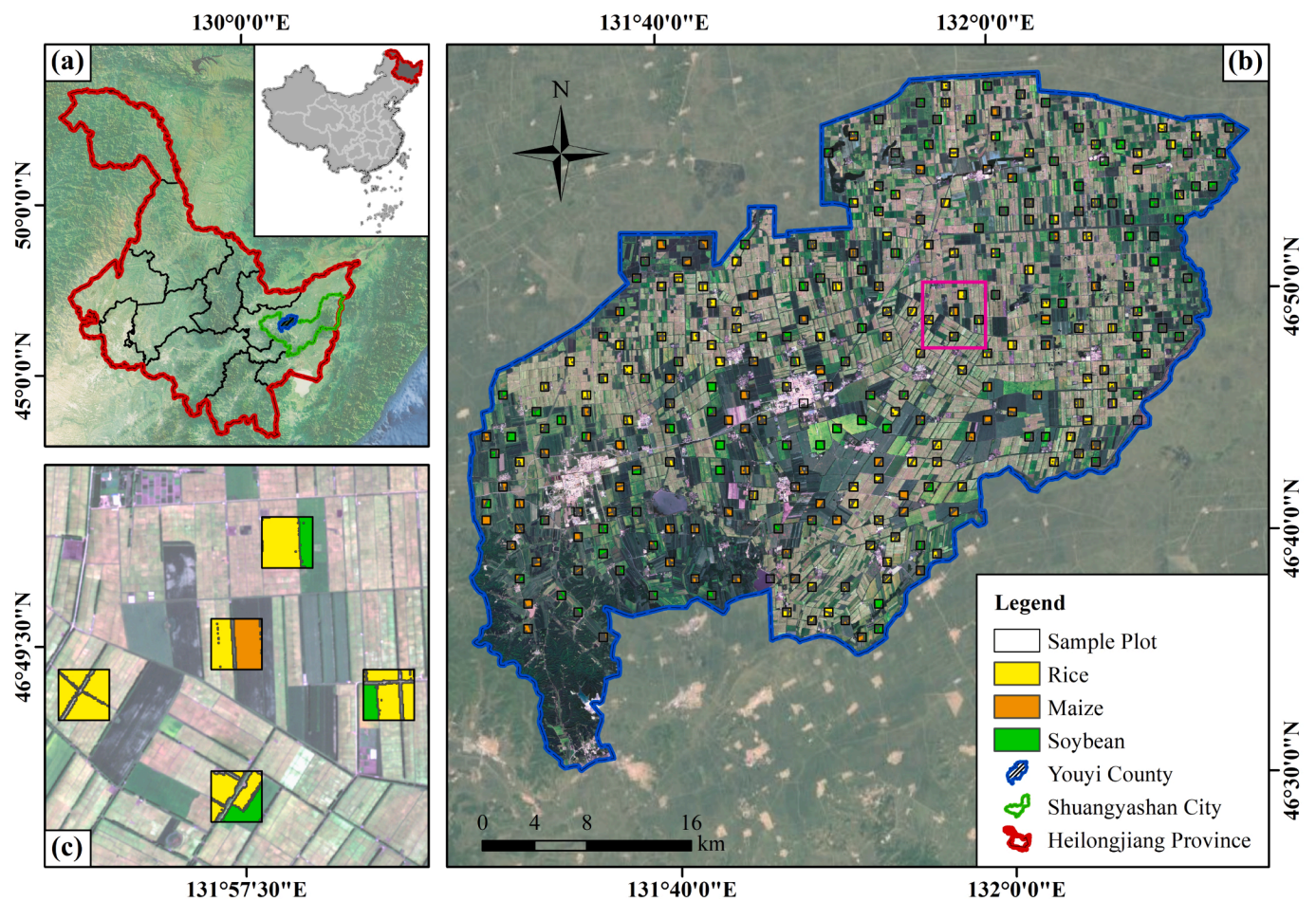


Fig. 1. Location and sample plot distribution of the study area. Specifically, (a) shows the locations of Youyi County, Shuangyashan City, and Heilongjiang Province; (b) represents the distribution map of the sample plots in the study area; and (c) is an enlarged view of the pink box shown in (b). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.1 °C, with an average yearly rainfall of 514 mm, an average sunlight duration of 2,557 h, and a frost-free period of 143 days, making it an exceptionally favorable region for agricultural production. The cultivation structure of this county is dominated by three crops, rice, corn, and soybeans, which are typically grown in contiguous fields. The phenological calendar for these three major crops is provided in Fig. 2.

2.2. Datasets and preprocessing

2.2.1. Ground survey data

To gain a comprehensive understanding of the crop planting

structure in Youyi County and the image features corresponding to various land objects, a ground survey and a data collection process were conducted from July to August of 2022. Utilizing a handheld GPS recorder, the type of each plot and its latitudinal and longitudinal information were marked at the center of the plot as sample points. A total of 268 plots with sizes of 640 m × 640 m were surveyed, from which 1269 sample points were collected; 490 were for rice, 389 were for corn, and 390 were for soybeans (Fig. 1). Based on these sample points, the plots were vectorized using ArcGIS 10.8, and a sample label dataset was created by using each plot as a unit. Detailed information about the labeled dataset was provided in Table 1.

Category	April		May		June			July			August			September			October		
	M	L	E	M	L	E	M	L	E	M	L	E	M	L	E	M	L	E	M
Rice	Seeding		Transplanting		Tillering			Jointing		Heading		Filling		Ripening					
Maize	Sowing		Seeding		Jointing			Tasseling		Flowering		Filling			Ripening				
Soybean			Sowing		Seeding		Flowering		Podding		Filling		Ripening						

Fig. 2. Phenological calendar of the major crops. In the figure, E, M, and L represent the early, middle, and late periods, respectively, of each month.

Table 1

Labeled dataset details.

Category	Number of Datasets	Number of Pixels				
		Rice	Maize	Soybean	Others	Total
Training Set	162	289,197	97,891	179,208	97,256	663,552
Validation Set	53	69,341	51,001	59,766	36,980	217,088
Test Set	53	61,417	58,021	61,540	36,110	217,088

2.2.2. Sentinel-2 image data

Compared to other satellites, Sentinel-2 has a higher data collection frequency and is the only satellite with three bands in the red-edge range, making it more effective for monitoring crop growth conditions. Based on the location of the study area, the phenological calendar of the three main crops (rice, maize, and soybean) as shown in Fig. 2, and cloud coverage conditions, four periods of available Level-1C Sentinel-2 data were retrieved and downloaded from the European Space Agency (<https://dataspace.copernicus.eu/>). Each period consists of three image scenes. Detailed information can be found in Table 2. Atmospheric correction was conducted using Sen2Cor version 02.11.00 to produce Level-2A products. In SNAP version 9.0.0, the Super Resolution plugin was used to interpolate the four 20-meter-spatial-resolution vegetation red-edge bands and the narrow near-infrared band to 10 m. Subsequent operations included mosaicking and cropping contemporaneous data.

3. Methods

3.1. Feature optimization

3.1.1. Classification feature

The features selected for crop classification in this study include three categories—spectral bands, vegetation indices, and textural features—providing a total of 168 features. Among them, spectral bands are essential model input features, while the other two categories serve as auxiliary features.

(1) Spectral Band

Due to the pigment and moisture content differences within leaves, various crops present distinct reflective features in remote sensing images. Therefore, distinguishing crops through the spectral bands of remote sensing data is a common and effective method for remote sensing-based crop classification. This study utilizes 32 spectral bands with a 10-meter spatial resolution derived from four S2 images.

(2) Vegetation Indices

Over the years, scholars in the field of remote sensing have

developed a variety of vegetation indices, which are widely used in crop remote sensing classification tasks. Based on the functions of these vegetation indices, we select 10 vegetation indices that can be calculated from the aforementioned spectral bands. Detailed information can be found in Table 3.

(3) Textural Features

The textural features of remote sensing images are essential spatial features and are mainly reflected by the frequency of spectral feature variations within a region. Different land cover types exhibit distinct textural features, so images with higher spatial resolutions can be distinguished based on their textural structure differences. In this study, we employ the gray-level co-occurrence matrix (GLCM) method for texture extraction purposes (Haralick et al., 1973) and select four effective texture features for calculation: homogeneity (HOM), dissimilarity (DISS), entropy (ENT), and the angular second moment (ASM). Their calculation formulas are as follows (Puissant et al., 2005):

$$\text{HOM} = \sum_{i,j=0}^{N_g-1} \frac{p(i,j)}{1 + |i - j|} \quad (1)$$

$$\text{DISS} = \sum_{i,j=0}^{N_g-1} p(i,j)|i - j| \quad (2)$$

$$\text{ENT} = \sum_{i,j=0}^{N_g-1} p(i,j) \log[p(i,j)] \quad (3)$$

$$\text{ASM} = \sum_{i,j=0}^{N_g-1} p(i,j)^2 \quad (4)$$

where i and j represent the coordinates in the gray-level co-occurrence matrix space; $p(i,j)$ represents the value of the (i,j) th entry in the gray-level co-occurrence matrix; and N_g represents the number of distinct gray levels in the quantized image.

The calculation of the four texture features is based on a 3×3 moving window, 1 pixel offset, and 64 Gy levels. The final value for each texture feature is obtained by averaging the feature values in the four directions: 0° , 45° , 90° , and 135° .

3.1.2. Feature importance evaluation

The Random Forest-SHAP (SHapley Additive exPlanations) algorithm, which combines the robust performance of the Random Forest classifier with the interpretability of SHAP, was used for feature selection. SHAP values provide a method for explaining machine learning model predictions, based on the Shapley values from cooperative game theory. By assigning a quantifiable importance value to each feature of the model, it explains the extent of each feature's contribution to the model's prediction. The calculation of SHAP values considers all possible combinations of features, ensuring a fair evaluation of each feature's contribution by assessing the changes in model predictions when a particular feature is included or excluded. The formula for calculating SHAP values is as follows (Lundberg and Lee, 2017):

$$\text{SHAP}_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \quad (5)$$

where i represents the feature being evaluated; F represents the set of all features; S represents a subset of features not containing i ; $|S|$ represents

Table 2

Sentinel-2 image data details.

Order Number	Satellite Platform	Shooting Date	Data Identification	Cloud Coverage (%)
T1	S2B	2022-05-09	N0400_R103_T52TFS	0.85
			N0400_R103_T52TGS	3.17
			N0400_R103_T52TGT	1.12
T2	S2B	2022-08-17	N0400_R103_T52TFS	14.21
			N0400_R103_T52TGS	4.55
			N0400_R103_T52TGT	0.00
T3	S2B	2022-09-09	N0400_R003_T52TFS	0.08
			N0400_R003_T52TGS	0.00
			N0400_R003_T52TGT	10.73
T4	S2B	2022-09-26	N0400_R103_T52TFS	0.00
			N0400_R103_T52TGS	0.00
			N0400_R103_T52TGT	0.00

Table 3

The selected vegetation indices (VIs) for the subsequent study.

VI	Full Name of VI	Calculation Formula	References
RVI	Ratio Vegetation Index	$\frac{R_{nir}}{R_{red}}$	(Jordan, 1969)
VDVI	Visible-Band Difference Vegetation Index	$\frac{2 \times R_{green} - R_{red} - R_{blue}}{2 \times R_{green} + R_{red} + R_{blue}}$	(Wang et al., 2015)
NDVI	Normalized Difference Vegetation Index	$\frac{R_{nir} - R_{red}}{R_{nir} + R_{red}}$	(Rouse et al., 1974)
GNDVI	Green Normalized Difference Vegetation Index	$\frac{R_{nir} - R_{green}}{R_{nir} + R_{green}}$	(Gitelson et al., 1996)
RDVI	Renormalized Difference Vegetation Index	$\frac{R_{nir} - R_{red}}{(R_{nir} + R_{red})^{0.5}}$	(Roujean and Breon, 1995)
ARVI	Atmospherically Resistant Vegetation Index	$\frac{(R_{nir} - R_{rb}) / (R_{nir} + R_{rb})}{R_{rb} = R_{red} + \gamma(R_{red} - R_{blue})}$	(Kaufman and Tanre, 1992)
SAVI	Soil-Adjusted Vegetation Index	$\frac{(1 + L)(R_{nir} - R_{red})}{R_{nir} + R_{red} + L}$	(Huete, 1988)
TVI	Triangular Vegetation Index	$0.5 \times [120 \times (R_{nir} - R_{green}) - 200 \times (R_{red} - R_{green})]$	(Broge and Leblanc, 2001)
NDRE1	Normalized Difference Red Edge Index 1	$\frac{R_{vire740} - R_{vire705}}{R_{vire740} + R_{vire705}}$	(Gitelson et al., 2003)
NDRE2	Normalized Difference Red Edge Index 2	$\frac{R_{vire783} - R_{vire705}}{R_{vire783} + R_{vire705}}$	(Gitelson et al., 2003)

* In the table, γ is taken as 1, and L is taken as 0.5.

the number of elements in set S ; $|F|$ represents the total number of all features; $f(S \cup \{i\})$ represents the model's prediction when including feature i and the features in set S ; $f(S)$ represents the model's prediction when including only the features in set S ; $f(S \cup \{i\}) - f(S)$ represents the difference between the two predictions and $S \subseteq F \setminus \{i\}$ represents that set S is one of all possible subsets obtained from set F by removing the specific feature i . A calculated $SHAP_i$ result that is greater than zero indicates that the corresponding feature has a positive influence on the model prediction, while a value less than zero indicates a negative impact.

3.2. Proposed model architecture

In this section, we focus on the CTANet model architecture proposed for small sample datasets, as illustrated in Fig. 3. It consists of two parts: a convolutional attention architecture (CAA) and a temporal attention architecture (TAA). The following sections provide detailed explanations of these two architectures.

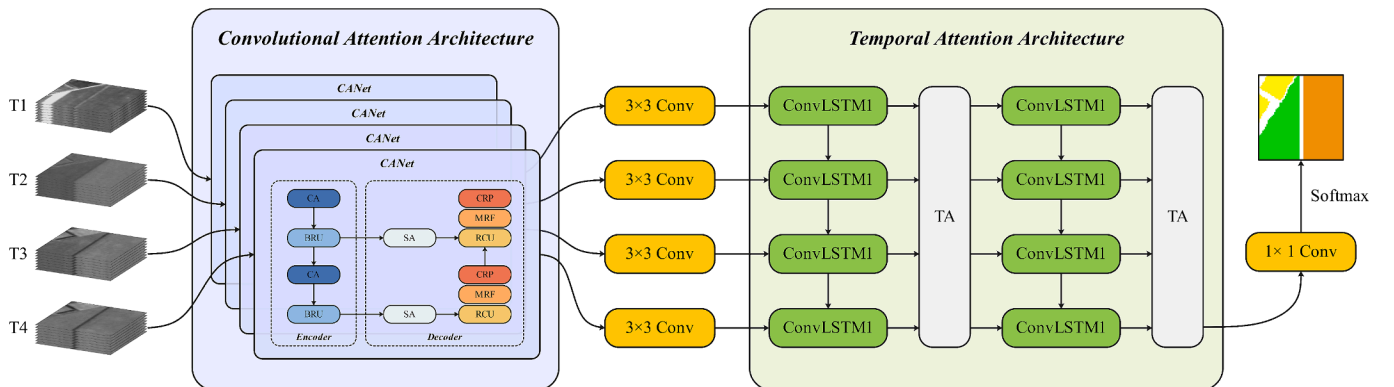
3.2.1. Convolutional attention architecture

The convolutional attention architecture consists of four parallel and identical convolutional attention networks (CANet). Due to the dataset's dimensions being only 64×64 pixels and the limited sample size, directly applying traditional four-scale architecture networks to such a small sample dataset often leads to ineffective feature extraction. Therefore, drawing on the design principles of RefineNet (Lin et al., 2017) and S^3 ANet (Hu et al., 2022), we construct a more lightweight

two-scale CANet. This design not only reduces the model's complexity but also retains sufficient feature extraction capability, making it better suited for training on small sample datasets. We introduce attention modules at each scale to optimize the feature processing strategies during the encoding and decoding stages. In the encoding stage, the model focuses on extracting key features from the input remote sensing images, capturing not only spatial features but also channel-level features. The latter reflects the importance of different spectral bands, which is crucial for crop classification. In the decoding stage, the model's task is to restore the high-dimensional feature representations to the spatial resolution of the input remote sensing images. The key to the decoding stage is to capture the spatial details of the input image to produce high-quality reconstructed images or accurate classification results. Therefore, we decompose the traditional CBAM (Convolutional Block Attention Module) (Woo et al., 2018) into two separate modules along the channel and spatial dimensions, integrating them into the encoder and decoder, respectively. The structure of the CANet is depicted in Fig. 4. The encoder is a novel backbone network composed of channel attention (CA) and bottleneck residual unit (BRU), while the decoder consists of spatial attention (SA) and incorporates the residual convolution unit (RCU), multi-resolution fusion (MRF), and chained residual pooling (CRP) from RefineNet.

1) Encoder

At each scale, a feature map F of size $H \times W \times C$ first passes through the CA module. Through both global maximum pooling (GMP) and global

**Fig. 3.** CTANet architecture diagram.

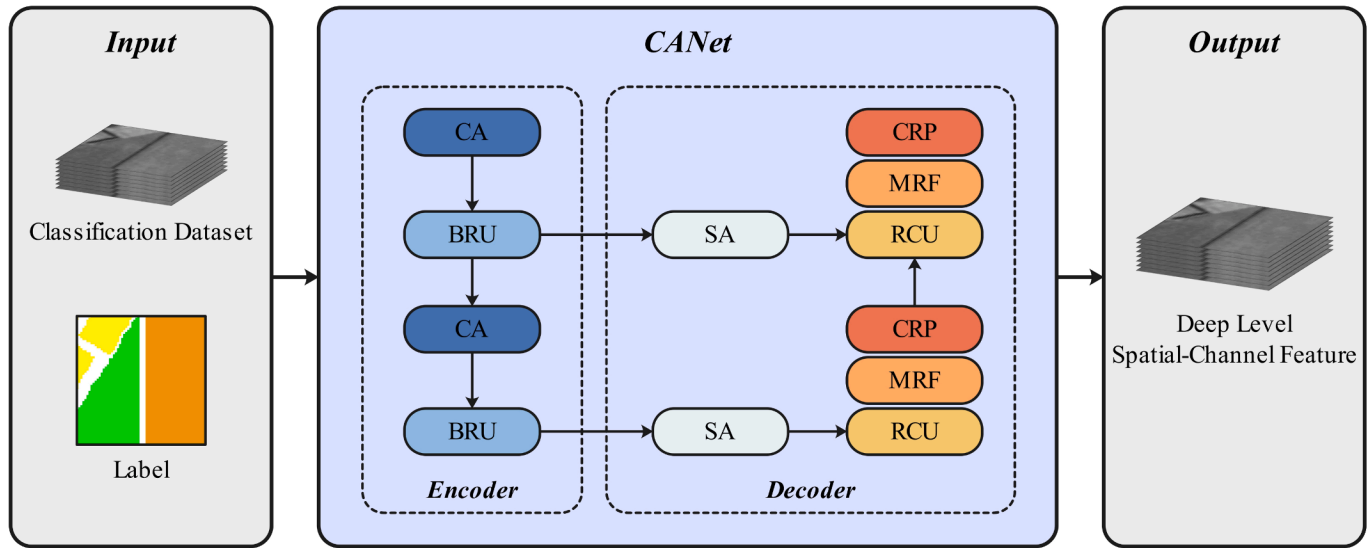


Fig. 4. CANet diagram.

average pooling (GAP) methods, two $1 \times 1 \times C$ channel features are acquired. These two channel features are subsequently fed into a shared two-layer neural network with a ReLU activation function. After summing the elements of the two output features, the weights for each channel are obtained through the sigmoid activation function and then multiplied by the input feature map F to obtain a channel attention map M_c (Fig. 5). The formula for doing so is as follows (Woo et al., 2018):

$$M_c(F) = \sigma(\text{SN}(\text{GMP}(F)) + \text{SN}(\text{GAP}(F))) \circ F \quad (6)$$

where $\circ$ represents the Hadamard product, which is an elementwise matrix multiplication operation and σ represents the sigmoid activation function.

Subsequently, the feature map enters the BRU, where the most significant difference from conventional residual modules is the introduction of a 1×1 convolution. The 1×1 convolution effectively reduces the feature dimensions without altering the spatial size of the feature map and also decreases the number of model parameters, which is particularly important for small sample datasets. Through this design, the BRU not only enhances the efficiency of feature extraction but also effectively prevents overfitting, which can be a concern when dealing with limited data. A diagram of the BRU is shown in Fig. 6 (He et al., 2016).

2) Decoder

The output feature maps at each scale of the encoder are fed into the decoder through the SA module. In the SA module, the feature map F is first subjected to average pooling and max pooling across its channels to

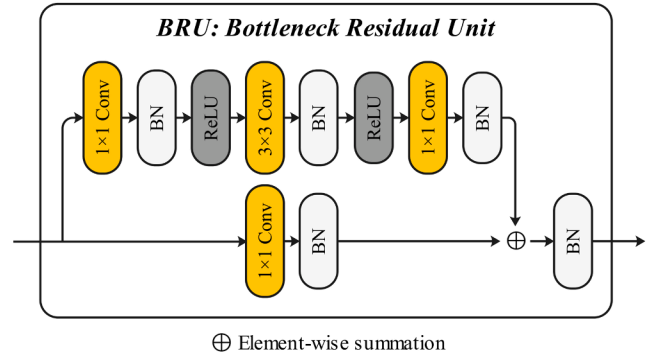


Fig. 6. Bottleneck residual unit diagram.

aggregate the channel information, compressing it dimensionally. Next, the two obtained channel features with sizes of $H \times W \times 1$ are concatenated. They are then passed through a convolutional layer with a 7×7 kernel size. Finally, a sigmoid activation function multiplies the weight coefficients with the input feature map F to produce a spatial attention map M_s (Fig. 7). The formula for this step is as follows (Woo et al., 2018):

$$M_s(F) = \sigma(\text{Conv}_{7 \times 7}([\text{GMP}(F); \text{GAP}(F)])) \circ F \quad (7)$$

where $\circ$ represents the Hadamard product, which is an elementwise matrix multiplication operation and σ represents the sigmoid activation function.

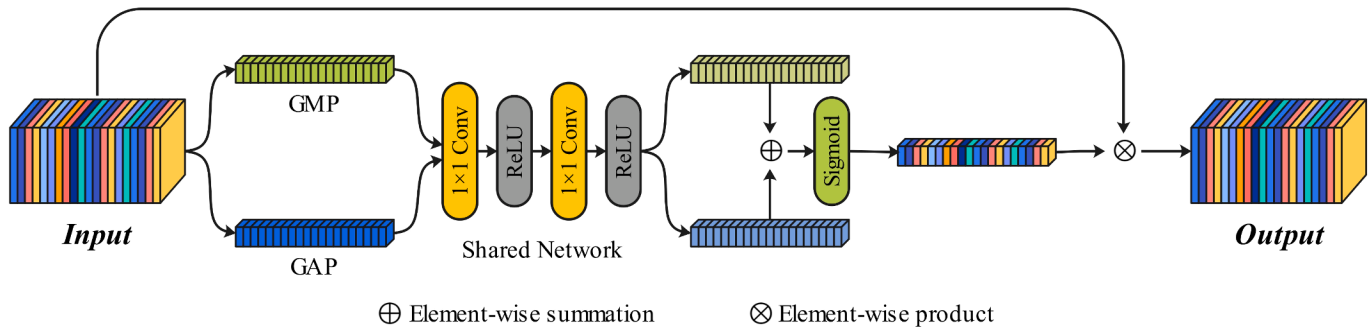


Fig. 5. Channel attention diagram.

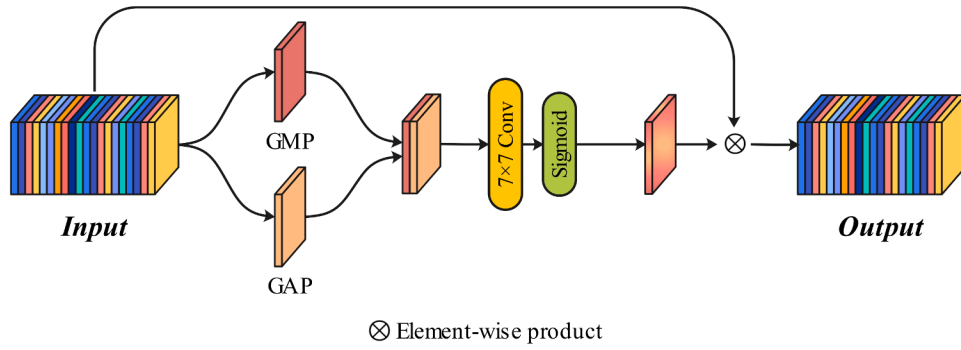


Fig. 7. Spatial attention diagram.

Next, the algorithm sequentially passes through RCU, MRF, and CRP modules. The RCU module is used to extract the residuals of the segmentation results at different resolutions and ultimately correct the original segmentation result at the corresponding resolution by adding them together, thus fine-tuning the weights of the segmented image. Each RCU consists of 2 ReLU layers and 2 convolution layers, as shown in Fig. 8(a) (Lin et al., 2017).

After obtaining the processed segmentation results at multiple resolutions, each result sequentially passes through a convolution layer and an upsampling layer, upscaling the input high-level feature map to the same size as that of the low-level semantic feature map produced by the BRU. This process creates a segmentation result map with a unified spatial resolution, which is subsequently summed. A schematic diagram of the MRF module is shown in Fig. 8(b) (Lin et al., 2017).

The CRP module consists of a residual structure, pooling layers, and convolution layers. Its function is to further optimize the segmentation results obtained from the previous two fusion steps using residual correction. The main purpose of combining pooling and convolution layers is to obtain residuals for correction. Instead of directly obtaining an independent residual from the corrected segmentation result, the residual from the previous level is used as the input of the residual learning module at the next level. This approach allows the subsequent modules to continue to perform in-depth learning based on the previous residuals, producing a better residual correction result. A schematic diagram of the CRP module is shown in Fig. 7(c) (Lin et al., 2017).

3.2.2. Temporal attention architecture

To delve deeper into the temporal-dimension features, we employ a dual-layer ConvLSTM network combined with temporal attention (TA) to construct a temporal attention architecture. ConvLSTM is an exemplary model for addressing temporal-spatial sequence prediction challenges and can be considered a variant of LSTM. Unlike LSTM, ConvLSTM transforms the connections between various gates and the input, as well as the connections between states, into convolutional operations. This means that the input at each time point can be a three-dimensional tensor, making this approach particularly suitable for handling remote sensing-based temporal-spatial sequence data. For the input X and each time step t , the structure of the ConvLSTM module is as shown in Fig. 9, and its specific operations can be represented as follows (Shi et al., 2015):

$$i_t = \sigma(W_{xi} * X_t + W_{hi} * H_{t-1} + W_{ci} \circ C_{t-1} + b_i) \quad (8)$$

$$f_t = \sigma(W_{xf} * X_t + W_{hf} * H_{t-1} + W_{cf} \circ C_{t-1} + b_f) \quad (9)$$

$$C_t = f_t \circ C_{t-1} + i_t \circ \text{Tanh}(W_{xc} * X_t + W_{hc} * H_{t-1} + b_c) \quad (10)$$

$$o_t = \sigma(W_{xo} * X_t + W_{ho} * H_{t-1} + W_{co} \circ C_t + b_o) \quad (11)$$

$$H_t = o_t \circ \text{Tanh}(C_t) \quad (12)$$

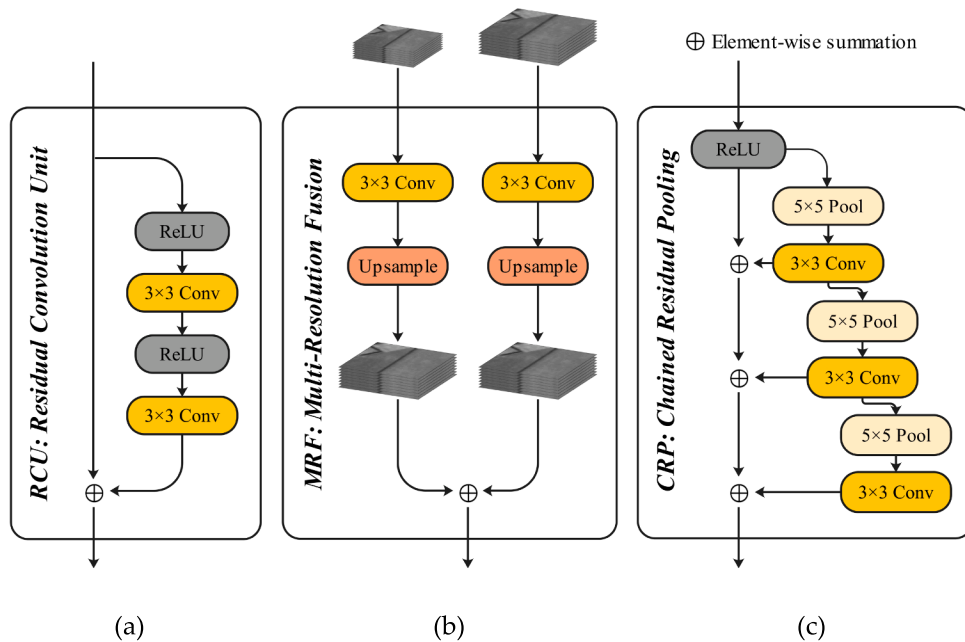


Fig. 8. Diagram of the RCU, MRF and CRP modules.

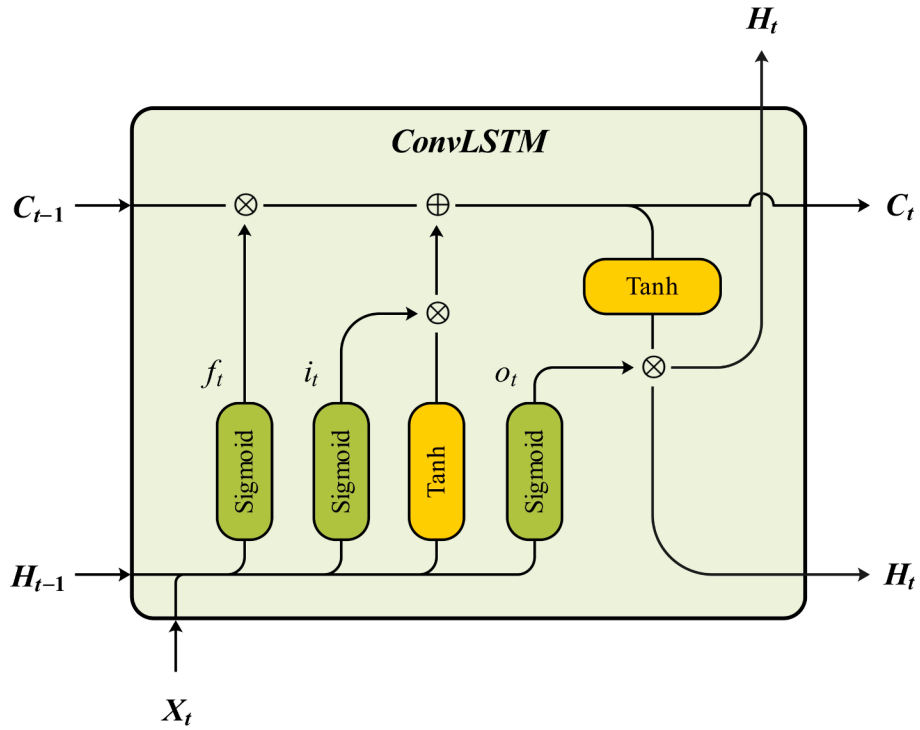


Fig. 9. ConvLSTM module diagram.

where $*$ represents the convolution operation; $\circ$ represents the Hadamard product, which is an elementwise matrix multiplication operation; W and b represent the weights and biases, respectively; σ and $\tanh$ represent the sigmoid activation function and the hyperbolic tangent (tanh) activation function, respectively; H represents the hidden state; i , f , and o represent the input gate, forget gate, and output gate, respectively; and C represents the cell state.

The purpose of the temporal attention layer is to assign different weights to the various time steps within the input feature sequence. To achieve this, we utilize a computationally efficient 1×1 convolutional kernel and a Sigmoid activation function to calculate the temporal attention weights. This design not only simplifies the model structure but also reduces the need for large-scale training data, allowing CTANet to maintain strong generalization capabilities even under small sample conditions. The resulting weighted feature sequence is represented as follows:

$$M_T(H_t) = \sigma(\text{Conv}_{1 \times 1}(H_t)) \circ H_t \quad (13)$$

where $\circ$ represents the Hadamard product, which is an elementwise matrix multiplication operation and σ represents the sigmoid activation function.

3.3. Experimental setup

Before model training, three data augmentation methods (horizontal flip, vertical flip, and diagonal mirroring) were applied to the training and validation sets of the three classification datasets to increase their quantity, thereby improving the model's generalization ability and performance. To comprehensively evaluate the performance of CTANet, we selected six different classification models (CANet, FarSeg, ConvLSTM, UNet, DeepLabv3+, and RefineNet) for comparison. The selection of these models is based on their widespread application and respective advantages in remote sensing image segmentation tasks. Since CTANet incorporates modules from RefineNet and ConvLSTM, it is both reasonable and necessary to include them as comparison models. UNet and DeepLabv3+ are classic models widely used in the remote

sensing field, known for their stability and efficiency. FarSeg is a recent model proposed for remote sensing tasks, demonstrating high segmentation performance. CANet is not only a crucial component of CTANet but also functions as an independent semantic segmentation model. Therefore, comparing these models allows us to comprehensively showcase the superiority and applicability of CTANet across different classification tasks. Additionally, based on three sets of classification datasets, we conducted a total of 21 experiments to systematically evaluate the performance improvements and advantages of CTANet. To ensure fairness, the four parallel CANet instances within CTANet were streamlined by reducing the number of convolutional kernels at each layer to one-fourth of those used in individual CANet instances. This adjustment aligned the parameter count of the CTANet model with that of other models for comparative evaluation. All experiments were developed and implemented using Python and the Keras framework on the Windows 11 operating system. Model training was conducted on an NVIDIA GeForce GTX 4070 Ti GPU and an Intel(R) Core(TM) i9-13900KF processor, using the adaptive moment estimation (Adam) optimization algorithm to adjust model parameters. The specific training parameters were set as follows: the maximum number of iterations was 200, the initial learning rate was $1e-4$, the batch size was 8, and the loss function was categorical cross-entropy. Each experiment was run independently 10 times, retaining the best result.

3.4. Accuracy assessment

In this study, six metrics were selected to quantitatively evaluate the classification performance of the models: the overall accuracy (OA), user accuracy (UA), producer accuracy (PA), F1 score (F1), intersection over union (IoU), and mean intersection over union (MIOU).

The confusion matrix is a representation of the correspondence between actual data and classification results in a matrix format. Its primary function is to count the number of correctly classified pixels and the number of misclassified pixels. The rows and columns of the matrix represent the pixel quantities of the sample label category and the classification category, respectively. The associated formula is as follows:

$$M = \begin{bmatrix} M_{11} & \cdots & M_{1j} \\ \vdots & \ddots & \vdots \\ M_{i1} & \cdots & M_{ij} \end{bmatrix} \quad (14)$$

where M_{ij} represents the number of pixels in class i that are recognized by the classifier as belonging to class j and M_{ii} represents the number of pixels belonging to correctly classified samples.

Based on the confusion matrix, the calculation formulas for the six evaluation metrics are as follows (Huang et al., 2021; Yi et al., 2022):

$$OA = \frac{\sum_{i=1}^n M_{ii}}{N} \times 100\% \quad (15)$$

$$N = \sum_{i=1}^n \sum_{j=1}^n M_{ij} \quad (16)$$

$$UA_i = \frac{M_{ii}}{\sum_{j=1}^n M_{ji}} \times 100\% \quad (17)$$

$$PA_i = \frac{M_{ii}}{\sum_{j=1}^n M_{ij}} \times 100\% \quad (18)$$

$$F1_i = \frac{2 \times UA \times PA}{(UA + PA)} \times 100\% \quad (19)$$

$$IoU_i = \frac{M_{ii}}{\sum_{j=1}^n M_{ji} + \sum_{j=1}^n M_{ij} - M_{ii}} \times 100\% \quad (20)$$

$$MIOU = \frac{\sum_{i=1}^n IoU_i}{n} \times 100\% \quad (21)$$

where n represents the number of sample categories; N represents the total number of sample pixels; and $\sum_{j=1}^n M_{ji}$ and $\sum_{j=1}^n M_{ij}$ represent the total pixel counts of the i th column and the i th row, respectively. UA_i , PA_i , $F1_i$, and IoU_i represent the user accuracy, producer accuracy, F1 score, and IoU of the i th category, respectively.

4. Results

4.1. Feature optimization results

To balance the number of features and the efficiency of the training process, we evaluated 168 auxiliary features, including vegetation indices and texture features, using the random forest-SHAP algorithm for importance determination. Considering the randomness of feature selection when establishing decision trees in random forests, the feature importance statistics also exhibit some randomness. Therefore, the modeling process was repeated 10 times, taking the average feature importance of the 10 models as the final classification feature importance level. The top 30 features, sorted from high to low importance, are shown in Fig. 10.

Fig. 10 displays the differences among the contributions of various features at different times to the classification of three crops: rice, maize, and soybeans. From a feature perspective, the same feature contributed differently to different categories at the same time. For instance, 0817_NDRE2 contributed the most to rice, followed by soybeans, and finally to maize. This reflects the high sensitivity of rice to the NDRE2 feature on August 17. The same feature contributed differently to the same crop type at different times. For example, NDRE2 contributed the least to rice on September 26. Additionally, among the top 30 features, those from 0817 and 0909 had the largest contributions, indicating the significant importance of these two time phases in distinguishing different crops. This corresponded to reality, as all three crops were in the grain-filling stage during this period; this a critical phase in their growth cycle where crops begin accumulating starch, protein, and other nutrients and are particularly responsive to environmental factors.

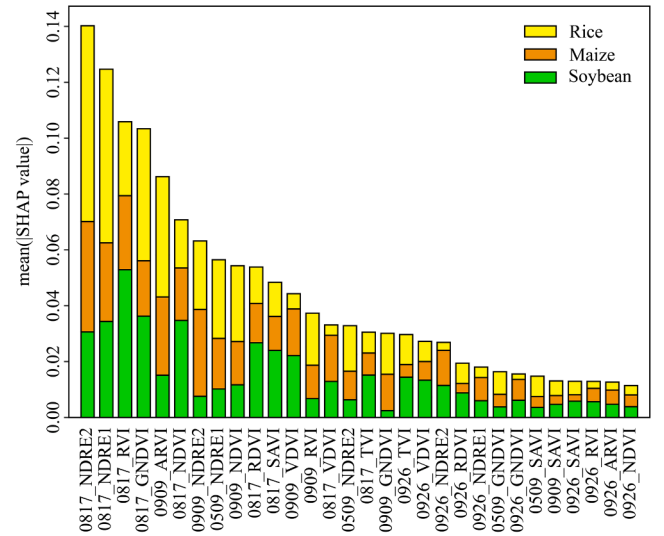


Fig. 10. Auxiliary feature importance evaluation results.

Notably, all 30 of these features are vegetation index features, with no textural features, confirming that vegetation indices capture more information related to the physiological and biochemical conditions associated with crop types than textural features.

To intuitively analyze the impacts of features on different crop categories, we constructed scatter plots of the SHAP values, as shown in Fig. 11. As shown in Fig. 11(a) and Fig. 11(c), the 0817_NDRE2 and 0817_NDRE1 features exhibited relatively extreme SHAP value distributions. These findings indicate that these features at this time point significantly contributed to predicting rice and soybeans, with considerable variability. In Fig. 11(b) and Fig. 11(c), the high values of most features are associated with positive SHAP values, indicating positive correlations. This means that increases in these feature values would increase the probability of the model predicting maize and soybeans. In contrast, most of the features in Fig. 11(a) exhibit negative correlations.

By using the average |SHAP value|, we further quantified the contributions of the features within each crop category. We ultimately selected 8 features as optimized auxiliary features: 0817_RVI, 0817_NDVI, 0817_GNDVI, 0817_RDVI, 0817_NDRE1, 0817_NDRE2, 0909_ARVI, and 0909_NDRE2. These features were categorized into a classification dataset consisting of spectral bands, forming a classification dataset with spectral bands and optimized auxiliary features. Therefore, we constructed three different classification datasets: a classification dataset with spectral bands (referred to as Dataset 1, containing 32 features), a classification dataset with spectral bands and auxiliary features (referred to as Dataset 2, containing 200 features), and a classification dataset with spectral bands and optimized auxiliary features (referred to as Dataset 3, containing 40 features). These datasets provided a foundation for our subsequent studies.

4.2. Classification results

Based on the confusion matrices, we calculated six evaluation metrics, the OA, UA, PA, F1, IoU, and MIOU, to further analyze the specific performance achieved by each model on different datasets (Table 4, Table 5 and Table 6). On dataset 1, CTANet exhibited excellent performance, especially for the maize and other categories, achieving the highest values for three metrics each: the UA (96.2 %), F1 score (96.2 %), and IoU (92.70 %) for maize and the PA (83.9 %), F1 score (84.5 %), and IoU (73.2 %) for the other category. Additionally, the highest F1 score and IoU were achieved for soybeans. FarSeg performed best in terms of identifying rice, reaching F1 score and IoU values of 94.7 % and 89.9 %, respectively, while the performance of CTANet slightly declined but remained competitive in terms of the UA. On Dataset 2, which has

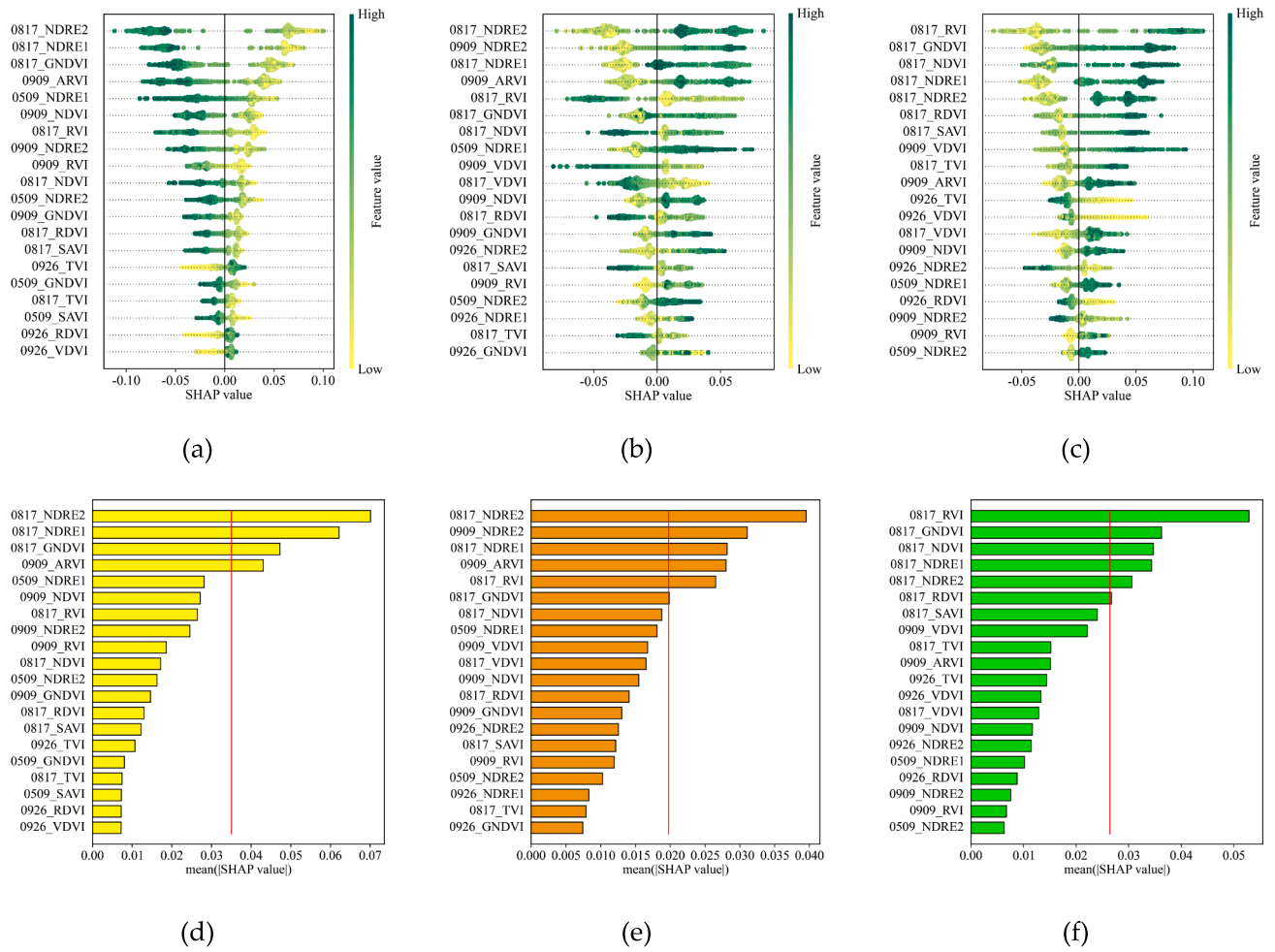


Fig. 11. Importance evaluation results. Specifically, (a), (b), and (c) show the distributions of the SHAP values produced by each feature for rice, maize, and soybean sample points, where each point represents a sample, with its color corresponding to the feature value and the horizontal axis corresponding to the impact of the SHAP value on the model output; (d), (e), and (f) show the average |SHAP value| for each feature of the sample rice, maize, and soybean points, respectively. The red lines in the graphs represent half of the maximum SHAP value for that crop type, and features located to the right of the red line are considered to have significant impacts on the prediction results obtained for that category. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 4

Comparison among the classification accuracies achieved by different models on Dataset 1. Specifically, the values that are both bolded and underlined represent the best results produced under the current evaluation metrics for that category.

Model	Rice				Maize				Soybean				Others				OA	MIoU
	UA	PA	F1	IoU	UA	PA	F1	IoU	UA	PA	F1	IoU	UA	PA	F1	IoU		
CTANet	94.7	94.1	94.4	89.4	96.2	96.2	96.2	92.7	93.6	94.9	94.3	89.1	85.1	83.9	84.5	73.2	93.2	86.1
CANet	92.7	95.7	94.2	89.1	95.2	97.0	96.1	92.5	94.0	94.2	94.1	88.8	88.5	80.6	84.4	73.0	93.1	85.8
FarSeg	93.8	95.6	94.7	89.9	94.8	97.3	96.0	92.3	93.6	94.8	94.2	89.0	88.0	79.4	83.5	71.7	93.1	85.7
ConvLSTM	93.7	95.1	94.4	89.3	94.4	96.0	95.2	90.8	93.3	95.1	94.2	89.0	86.8	79.5	83.0	71.0	92.7	85.0
UNet	94.6	94.5	94.5	89.6	94.4	96.4	95.4	91.2	93.3	94.5	93.9	88.5	84.8	80.2	82.4	70.1	92.6	84.9
DeepLabv3+	92.3	94.6	93.5	87.7	93.8	96.8	95.3	91.0	93.8	92.7	93.2	87.3	85.9	79.5	82.5	70.3	92.1	84.1
RefineNet	92.2	92.9	92.6	86.2	95.0	94.9	94.9	90.3	94.4	90.8	92.6	86.2	77.8	81.9	79.8	66.4	91.0	82.3

the most features, the overall performances of all the models decreased compared to those attained on Dataset 1; however, CTANet still performed well in terms of recognizing the maize and other categories. The UNet achieved the highest recognition accuracy for rice and soybeans, with F1 scores of 94.6 % and 89.8 %, respectively, and IoU scores of 94.0 % and 88.8 %, respectively, ranking second in the OA and MIoU metrics (just behind CTANet). The performances of all the models on Dataset 3 were particularly noteworthy, as they achieved their highest OA and MIoU values. CTANet achieved the best performance on this

feature-optimized dataset, with overall accuracy and MIoU values reaching 93.9 % and 87.5 %, respectively. In terms of category recognition, CTANet was the most accurate method for the rice and other categories, while CANet and FarSeg were the most accurate methods for soybeans and maize, respectively. Notably, CTANet consistently ranked among the top three in recognition performance across all categories, exhibiting outstanding overall performance. Specifically, the F1 scores for rice, corn, and soybeans were 95.6 %, 95.7 %, and 94.7 %, respectively, while the IoU scores were 91.6 %, 91.7 %, and 89.9 %, respectively.

Table 5

Comparison among the classification accuracies achieved by different models on Dataset 2. Specifically, the values that are both bolded and underlined represent the best results produced under the current evaluation metrics for that category.

Model	Rice				Maize				Soybean				Others				OA	MIoU
	UA	PA	F1	IoU	UA	PA	F1	IoU	UA	PA	F1	IoU	UA	PA	F1	IoU		
CTANet	92.0	95.9	93.9	88.5	95.6	<u>96.3</u>	<u>96.0</u>	<u>92.3</u>	<u>95.0</u>	92.7	93.8	88.4	86.0	82.2	<u>84.1</u>	<u>72.5</u>	<u>92.8</u>	<u>85.4</u>
CANet	90.5	<u>96.2</u>	93.3	87.4	94.3	95.0	94.6	89.8	93.5	93.1	93.3	87.5	<u>86.6</u>	77.1	81.6	68.9	91.8	83.4
FarSeg	91.6	95.1	93.3	87.5	95.5	95.6	95.5	91.5	93.6	93.5	93.6	87.9	86.6	80.9	83.6	71.9	92.4	84.7
ConvLSTM	93.4	93.5	93.5	87.7	95.2	96.0	95.6	91.6	93.8	93.6	93.7	88.1	83.3	82.2	82.8	70.6	92.3	84.5
UNet	<u>95.0</u>	94.2	<u>94.6</u>	<u>89.8</u>	<u>95.8</u>	94.1	95.0	90.4	93.8	<u>94.3</u>	<u>94.0</u>	<u>88.8</u>	81.8	<u>84.5</u>	83.1	71.1	92.6	85.0
DeepLabv3+	90.7	95.4	93.0	86.9	94.6	95.9	95.2	90.9	93.9	92.7	93.3	87.4	85.8	78.3	81.9	69.3	91.9	83.6
RefineNet	90.7	94.4	92.5	86.0	95.1	93.7	94.4	89.4	93.7	90.5	92.1	85.3	79.6	80.6	80.1	66.8	90.8	81.9

Table 6

Comparison among the classification accuracies achieved by different models on Dataset 3. Specifically, the values that are both bolded and underlined represent the best results produced under the current evaluation metrics for that category.

Model	Rice				Maize				Soybean				Others				OA	MIoU
	UA	PA	F1	IoU	UA	PA	F1	IoU	UA	PA	F1	IoU	UA	PA	F1	IoU		
CTANet	95.1	<u>96.2</u>	<u>95.6</u>	<u>91.6</u>	<u>95.8</u>	95.6	95.7	91.7	95.2	94.2	94.7	89.9	86.8	<u>86.8</u>	<u>86.8</u>	<u>76.7</u>	<u>93.9</u>	<u>87.5</u>
CANet	<u>95.5</u>	93.9	94.7	90.0	93.9	<u>97.3</u>	95.5	91.5	<u>95.4</u>	94.4	<u>94.9</u>	<u>90.2</u>	84.7	83.6	84.1	72.6	93.2	86.1
FarSeg	94.2	95.1	94.7	89.9	94.9	96.8	<u>95.9</u>	<u>92.0</u>	94.7	<u>94.7</u>	94.7	90.0	85.7	81.5	83.6	71.8	93.2	85.9
ConvLSTM	93.8	95.9	94.8	90.1	94.7	96.2	95.4	91.3	94.7	94.2	94.5	89.5	86.7	82.1	84.4	72.9	93.2	86.0
UNet	92.3	95.8	94.0	88.7	95.2	96.3	95.7	91.8	93.6	94.6	94.1	88.8	<u>87.5</u>	78.7	82.8	70.7	92.7	85.0
DeepLabv3+	94.3	94.3	94.3	89.3	94.5	95.9	95.2	90.8	95.1	92.8	94.0	88.6	83.3	84.7	84.0	72.4	92.7	85.3
RefineNet	91.6	95.2	93.4	87.6	92.8	94.8	93.8	88.3	92.3	93.5	92.9	86.7	86.5	75.6	80.7	67.7	91.4	82.6

respectively. Additionally, in CANet, the UA for rice and the PA for corn achieved outstanding results. Similarly, in FarSeg, the PA for soybeans and in UNet, the UA for other categories also demonstrated excellent performance. A longitudinal analysis of the datasets from the model perspective revealed that each model performed best on Dataset 3 and worst on Dataset 2.

Further, we conducted a significance analysis of the seven models and 18 metrics within each dataset using the Friedman test (Table 7). The Friedman test statistics for the three datasets were 48.03, 59.18, and 52.26, respectively, with corresponding p-values of $1.17\text{E-}08$, $6.59\text{E-}11$, and $1.66\text{E-}09$. These p-values are significantly lower than the conventional significance level of 0.05, indicating that the performance differences of the CTANet model across different datasets are statistically significant. The results presented in Table 4, Table 5, Table 6 and Table 7 not only demonstrate the performance variations of different models when handling diverse datasets and feature combinations but also highlight the stability of CTANet's performance in multi-crop classification tasks.

Subsequently, we conducted a detailed comparison among the classification results obtained on Dataset 3 (Fig. 12), where the performance differences among the various models were especially evident under different scenarios. In terms of background connectivity, both CTANet and CANet handled it well. However, CANet's strong detail recognition capability led to the occurrence of the isolated island effect and some misclassification in edge regions. These phenomena were also observed in the classification results of ConvLSTM, UNet, DeepLabv3+, and RefineNet. In region (e), CTANet and FarSeg accurately delineated the boundary of the rice field on the left side, while CANet performed the worst in boundary handling. When classifying large-scale crops, FarSeg, ConvLSTM, UNet, DeepLabv3+, and RefineNet demonstrated stable performance, though they showed some shortcomings in handling complex boundaries. In some areas, the background (others) was

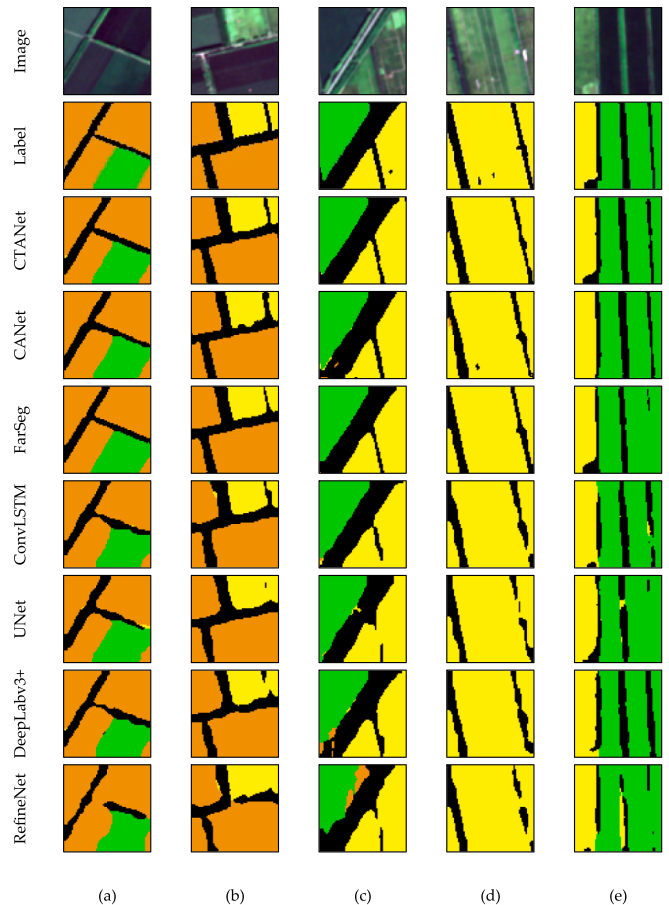


Fig. 12. Comparison among classification results obtained on Dataset 3 using different classification models. Specifically, in (a), (b), (c), (d), and (e), each subplot from top to bottom successively shows the images, labels, and classification results produced for the same region by CTANet, CANet, FarSeg, ConvLSTM, UNet, DeepLabv3+, and RefineNet, respectively.

Table 7

Significance analysis of CTANet based on Friedman test.

Metric	Dataset 1	Dataset 2	Dataset 3
Statistic	48.03	59.18	52.26
P-value	$1.17\text{E-}08$	$6.59\text{E-}11$	$1.66\text{E-}09$

erroneously labeled as crops, reflecting the models' insufficient sensitivity to small-scale features. In contrast, CTANet and CANet exhibited strong advantages in boundary processing. Additionally, none of the models effectively distinguished the island-like background from the rice in regions (c) and (d). Overall, CTANet had the lowest misclassification rate, with its classification results being the closest to the manually annotated ground truth.

Finally, we applied the CTANet model, trained on Dataset 3, to the entire study area, and after stitching, obtained the spatial distribution map of major crops produced in Youyi County for the year 2022 (Fig. 13). It is evident that rice is the dominant crop in Youyi County, with extensive and continuous planting areas, indicating a large-scale, concentrated planting pattern. Corn is primarily planted in the central region of the county, forming a distinct planting center. Soybeans, on the other hand, exhibit a dispersed planting pattern, with smaller and scattered planting areas throughout the study area. This distribution map not only reveals the crop planting patterns in Youyi County but also provides a reference for future agricultural planning and management.

5. Discussion

5.1. Analysis of training efficiency

Fig. 14 shows the validation accuracy and loss curves yielded by CTANet and six comparative models on three datasets during the model training process. In Dataset 1, we observe that CANet demonstrates a significant improvement in accuracy during the early stages of training, achieving over 90 % validation accuracy within just 25 epochs. Meanwhile, compared to other classification models, UNet exhibits greater volatility. When experimenting with Dataset 2, the highest validation accuracy for all models decreases, and the fluctuations become more pronounced. This phenomenon indicates that a large number of classification features can introduce more noise and irrelevant information, thereby interfering with model training. However, CTANet still demonstrates good stability. In Dataset 3, although the number of features is reduced, these features positively contribute to the prediction of each crop category. CTANet shows significant advantages on this dataset, achieving higher validation accuracy and lower validation loss. Overall, the performance of CTANet across different feature sets demonstrates its

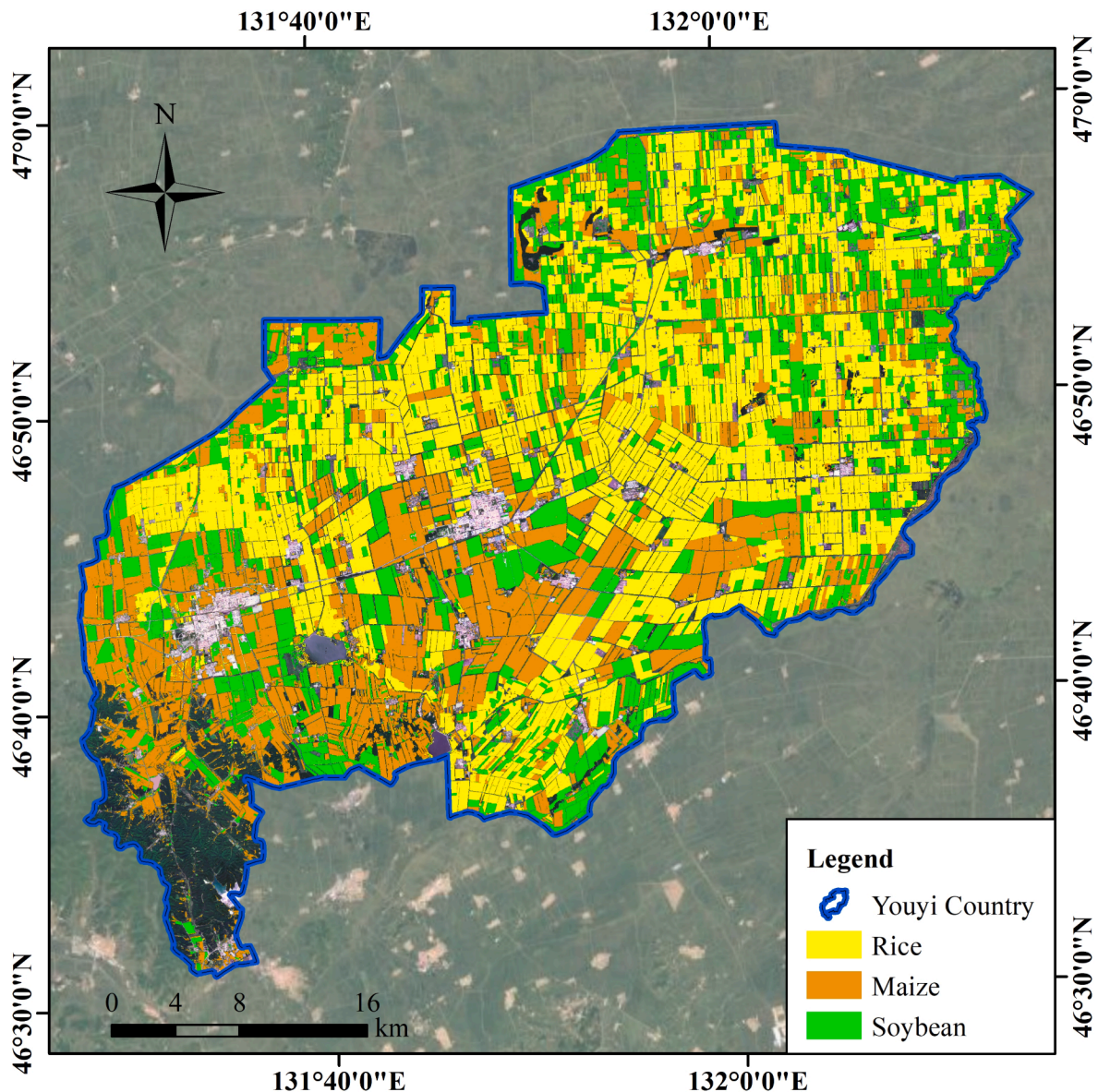


Fig. 13. Spatial distribution map of the major crops produced in Youyi County for the year 2022.

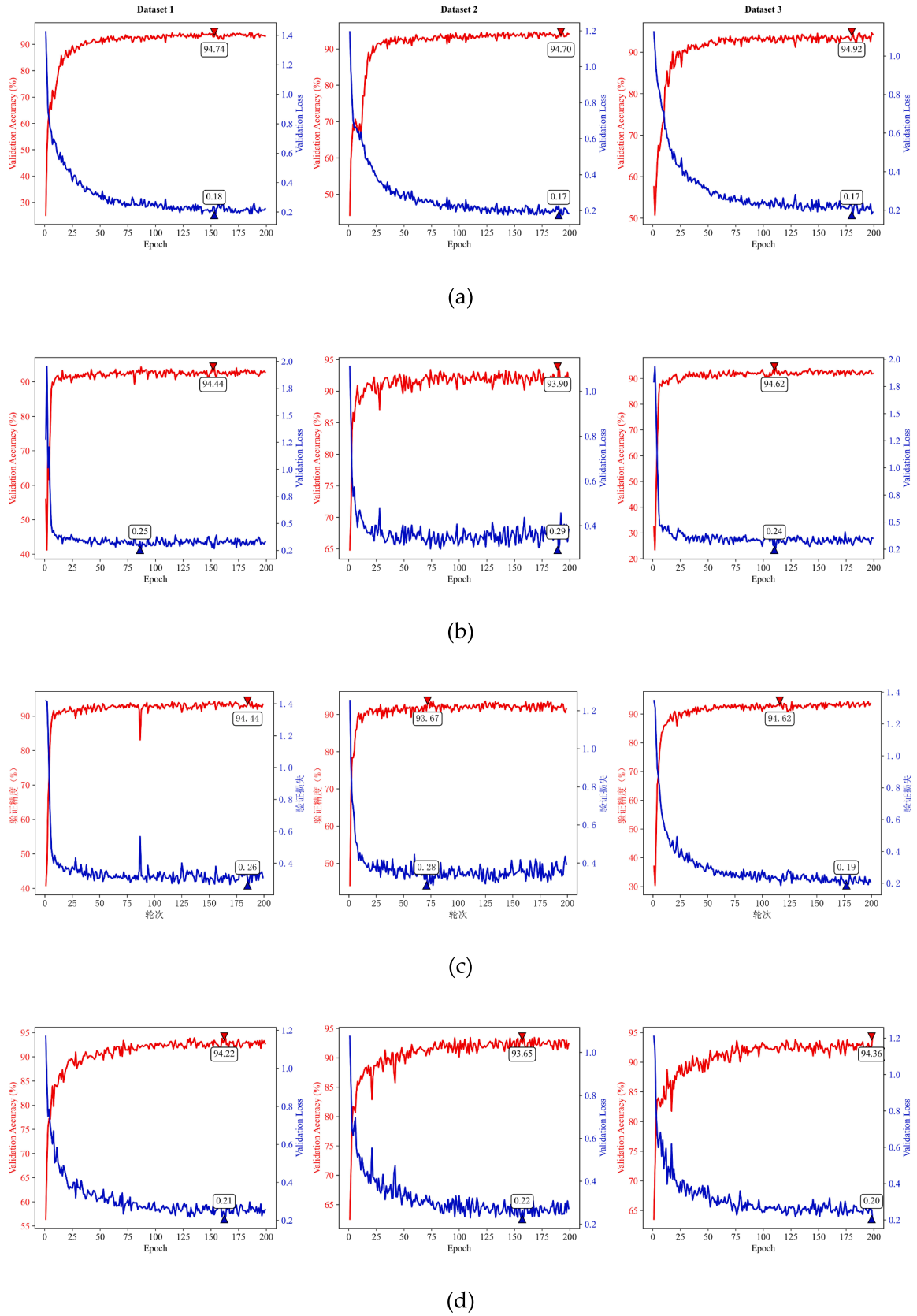


Fig. 14. Validation accuracy and loss curves produced by CTANet and five comparison models. Specifically, (a), (b), (c), (d), (e), (f) and (g) correspond to the validation accuracy and loss curves yielded by CTANet, CANet, FarSeg, ConvLSTM, UNet, DeepLabv3+, and RefineNet, respectively, based on the three datasets. The subplots from left to right represent Dataset 1, Dataset 2, and Dataset 3. Inside each subplot, red markers indicate the highest validation accuracy, with the maximum value noted in the box, while blue markers denote the lowest validation accuracy, with the minimum loss value noted in the box. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

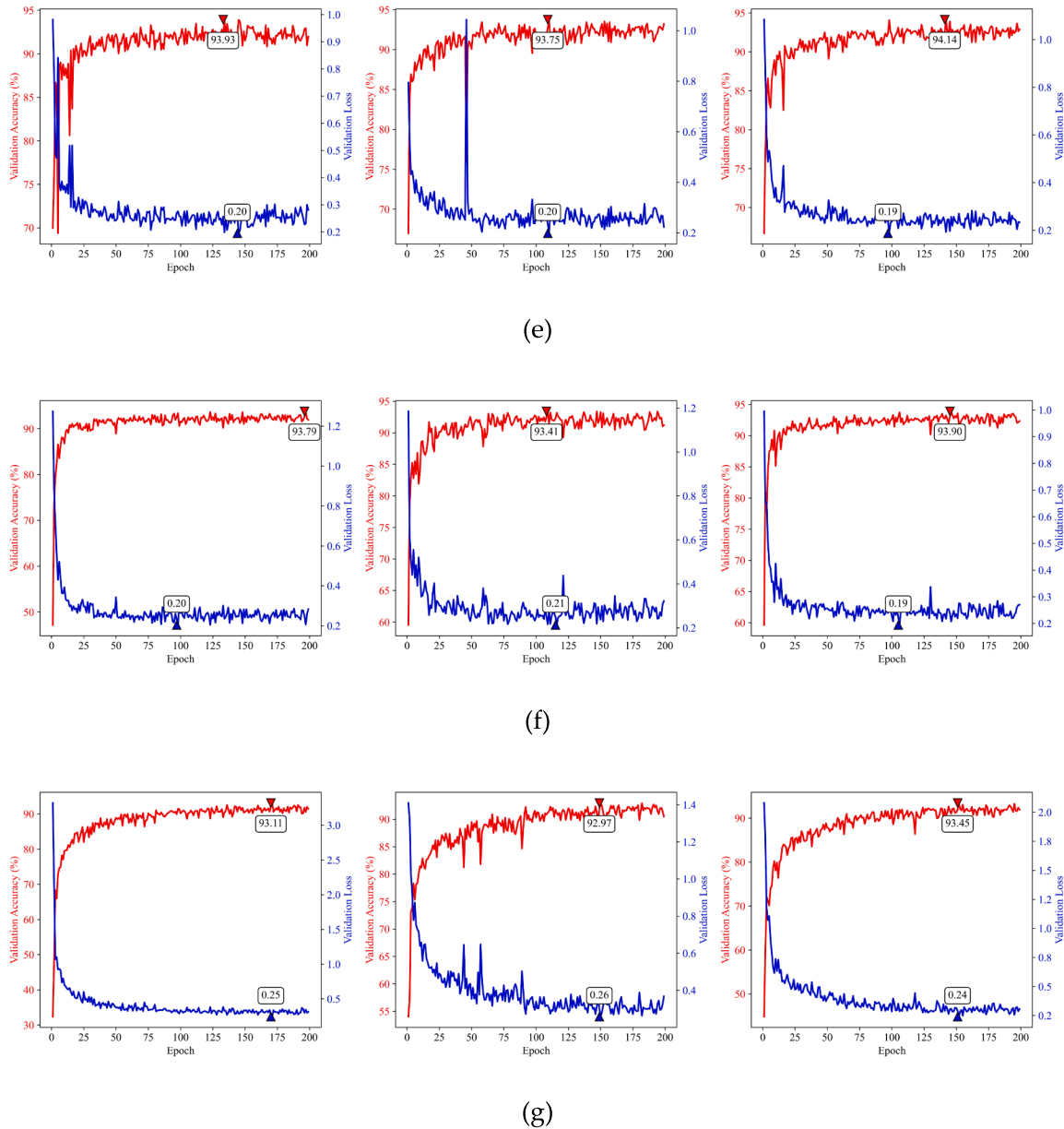


Fig. 14. (continued).

superiority and robustness in handling diverse data.

5.2. Analysis of the advantages of Dataset 3

In traditional methods, spectral bands, texture, and vegetation indices are usually selected together as features. However, in our method, spectral bands are treated as necessary inputs, and feature selection is performed only on auxiliary features (such as texture and vegetation indices). The advantage of this approach is that it ensures the completeness of essential information. Spectral bands contain crucial spectral information about crops, which is key for crop classification. Therefore, by treating spectral bands as necessary inputs, we ensure that the classification model always retains this critical information, unaffected by the feature selection process, thereby avoiding the potential loss of important spectral information during feature selection. At the same time, selecting only auxiliary features for feature selection reduces the computational complexity of the feature selection process. Since spectral bands are necessary inputs and do not participate in the selection process, this approach reduces computational costs and increases

model training efficiency. Through feature selection, the model can identify the most representative auxiliary features (such as specific textures and vegetation indices), which can provide additional discriminative information and further improve classification performance. This method can more effectively utilize auxiliary features to enhance classification outcomes.

Compared to the dataset using only spectral bands (Dataset 1), although such a dataset is computationally simple and ensures the inclusion of all essential spectral information, its classification performance may be limited due to the lack of auxiliary feature information, especially in complex environments or when spectral differences between different crops are minimal. In contrast, Dataset 3 not only includes spectral bands but also incorporates the most critical auxiliary features through feature selection, providing more comprehensive feature information, thus improving classification accuracy and robustness. Compared to the dataset using spectral bands plus all auxiliary features (Dataset 2), although this method includes all available features and potentially a large amount of information, it also introduces more noise and irrelevant information, which can interfere

with model training. In contrast, Dataset 3, through feature selection, eliminates redundant and useless auxiliary features, retaining only those features that significantly contribute to classification. This method not only reduces computational complexity but also enhances the model's generalization ability and classification performance.

5.3. Ablation study concerning the attention module in CTANet

Fig. 15 shows the results of ablation studies performed regarding the spatial, channel, and temporal attention modules in CTANet on three datasets under the same hyperparameters. By observing the bar chart, it is visually evident that different modules contribute uniquely to the enhancement of the model's performance. The baseline model results demonstrate the fundamental performance level across various datasets without incorporating any specialized attention mechanisms. With the introduction of spatial, channel, and temporal attention mechanisms, improvements in accuracy are observed across all three classification datasets, with temporal attention contributing the most significantly to performance enhancement. This underscores the importance of capturing temporal dynamic features for improving crop classification performance. When combining two mechanisms (spatial + channel, spatial + temporal, and channel + temporal), the channel + temporal combination yielded the best performance enhancement. This combination effectively integrates the strengths of both channel and temporal attention mechanisms, enhancing the model's ability to understand inter-feature relationships and temporal variations, thereby making the model more efficient in handling data with complex channel and temporal information. Furthermore, the study found that the comprehensive use of spatial, channel, and temporal attention mechanisms led to performance improvements surpassing those achieved by combining any single or any two mechanisms, achieving the highest overall accuracy (OA) across six models. This result was validated on all three classification datasets. The ablation experiment results not only confirmed the effectiveness of multi-dimensional feature integration but also highlighted the importance of integrating different types of attention

modules in the design of deep learning models.

5.4. Transfer learning of CTANet

To verify the adaptability and generalization performance of CTANet across different years and regions, we supplemented the classification datasets for Youyi County in 2021 and Yilan County in 2022 (Datasets 4 and 5). These two datasets have the same feature combinations and production requirements as dataset 3 but are only 25 % of its size, including 41 training images, 13 validation images, and 13 test images. Similarly, we applied three data augmentation methods to increase the number of training and validation images in these two classification datasets. Subsequently, we fine-tuned the CTANet model, which was initially trained on dataset 3, using the augmented datasets. The performance of CTANet on Datasets 4 and 5 is shown in Table 8. In both datasets, the OA exceeded 90 %, and the MIoU exceeded 80 %. For dataset 4, the UA, PA, and F1-score for rice, maize, and soybean all exceeded 92 %. In dataset 5, the F1-scores for rice, maize, and soybean all exceeded 91 %. This indicates that the CTANet model, after data augmentation and fine-tuning, exhibits good generalization ability across different years and regions. The two fine-tuned CTANet models were applied to the respective study areas, resulting in the spatial distribution maps of major crops in Youyi County for 2021 (Fig. 16(a)) and Yilan County for 2022 (Fig. 16(b)).

6. Conclusions

In this paper, a fine crop classification model based on multitemporal Sentinel-2 images, CTANet, is proposed. It comprises a convolutional attention architecture (CAA) and a temporal attention architecture (TAA), incorporating spatial attention modules, channel attention modules and temporal attention modules. These modules adaptively weight each pixel, channel and temporal phase of the given feature map to mitigate the intraclass spatial heterogeneity, spectral variability and temporal variability of crops. In comparison with models such as CANet,

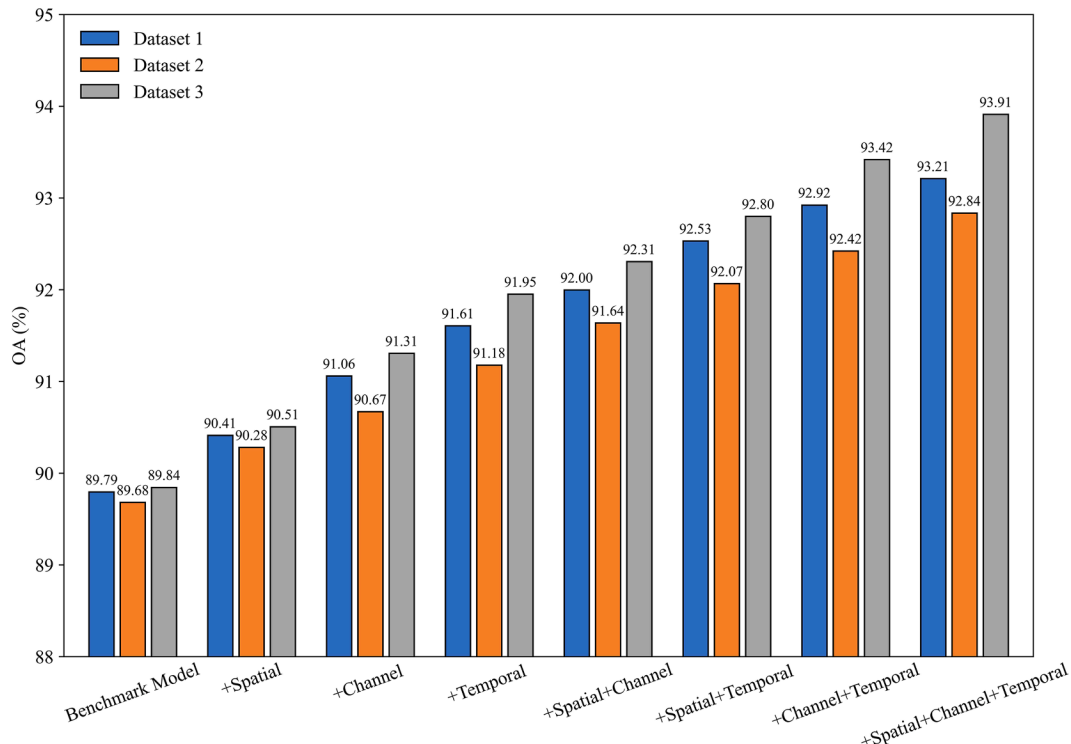


Fig. 15. Ablation study conducted for the attention module in CTANet on three datasets. Specifically, the baseline model means that no attention modules are used. Spatial, Channel, and Temporal represent the spatial, channel, and temporal attention modules, respectively.

Table 8
Comparison of classification accuracies achieved by CTANet on different datasets.

Dataset	Rice				Maize				Soybean				Others				OA	MIoU
	UA	PA	F1	IoU	UA	PA	F1	IoU	UA	PA	F1	IoU	UA	PA	F1	IoU		
Dataset 4	93.4	95.8	94.6	89.8	93.2	93.2	93.2	87.3	92.1	92.6	92.4	85.8	84.5	77.1	80.6	67.6	91.9	82.6
Dataset 5	89.1	95.8	92.3	85.8	92.9	89.8	91.3	84.1	90.6	92.5	91.6	84.4	85.0	76.1	80.3	67.1	90.1	80.3

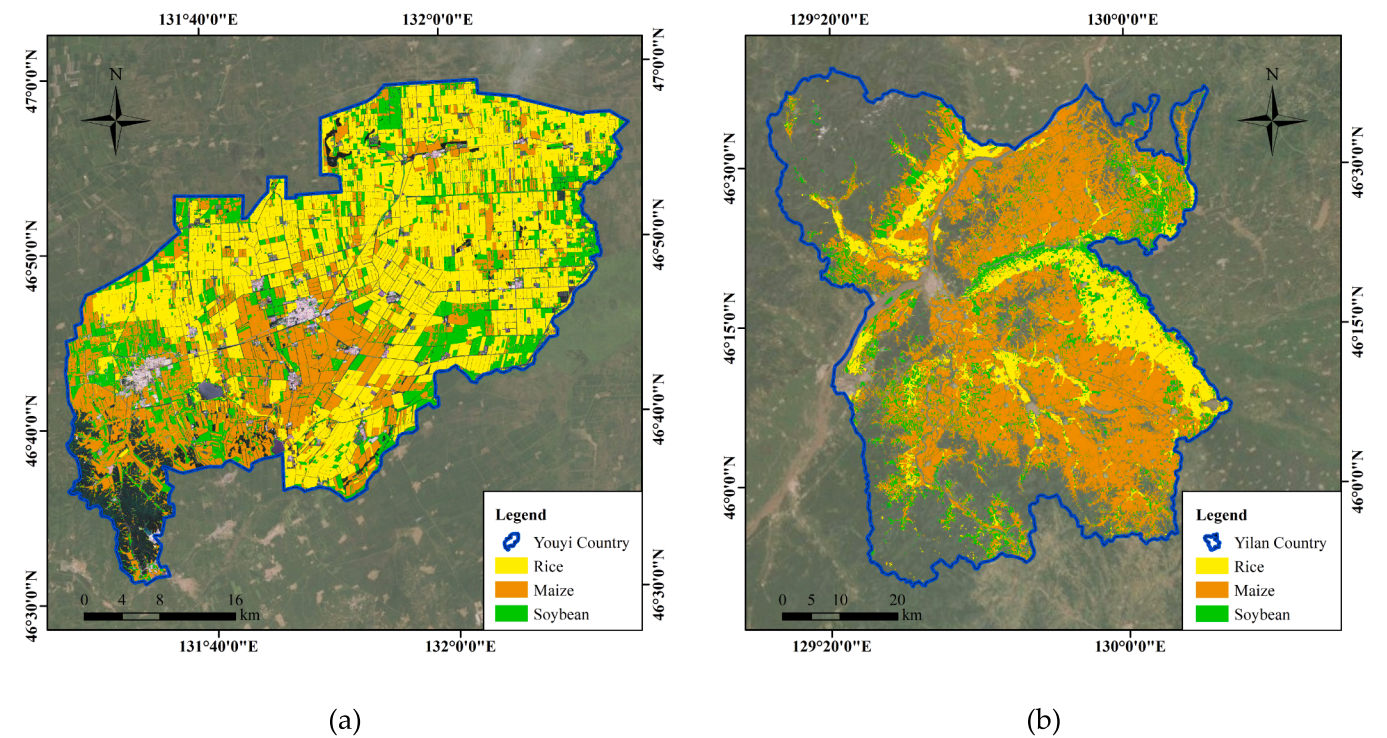


Fig. 16. Spatial distribution maps of major crops based on transfer learning. Specifically, (a) shows the spatial distribution of major crops in Youyi County for the year 2021, and (b) shows the spatial distribution of major crops in Yilan County for the year 2022.

FarSeg, ConvLSTM, UNet, DeepLabv3+, and RefineNet, CTANet had significant advantages in multicrop classification tasks, especially on the classification dataset consisting of spectral bands and optimized auxiliary features. During training, it achieved higher validation accuracy and lower validation loss than those of the other methods, and during testing, it attained the highest overall accuracy (93.9 %) and MIoU (87.5 %). When identifying rice, maize, and soybeans, the F1 scores reached 95.6 %, 95.7 %, and 94.7 %, and the IoU scores were 91.6 %, 91.7 %, and 89.9 %, respectively, significantly exceeding those of some commonly used deep learning models. Moreover, the introduction of spatial attention modules, channel attention modules, and temporal attention modules played a key role in enhancing the crop classification accuracy of the model. Their combined effect significantly improved classification performance of the proposed approach, surpassing that of any single module or any combination of two modules. These experiments indicated that CTANet not only demonstrated innovation at the technical level but also exhibited strong potential and practical value for use in real-world crop classification applications.

CRedit authorship contribution statement

Tengfei Qu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hong Wang:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Xiaobing Li:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition,

Conceptualization. **Dingsheng Luo:** Visualization, Validation, Software, Investigation, Formal analysis. **Yalei Yang:** Writing – review & editing, Investigation. **Jiahao Liu:** Conceptualization. **Yao Zhang:** Validation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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